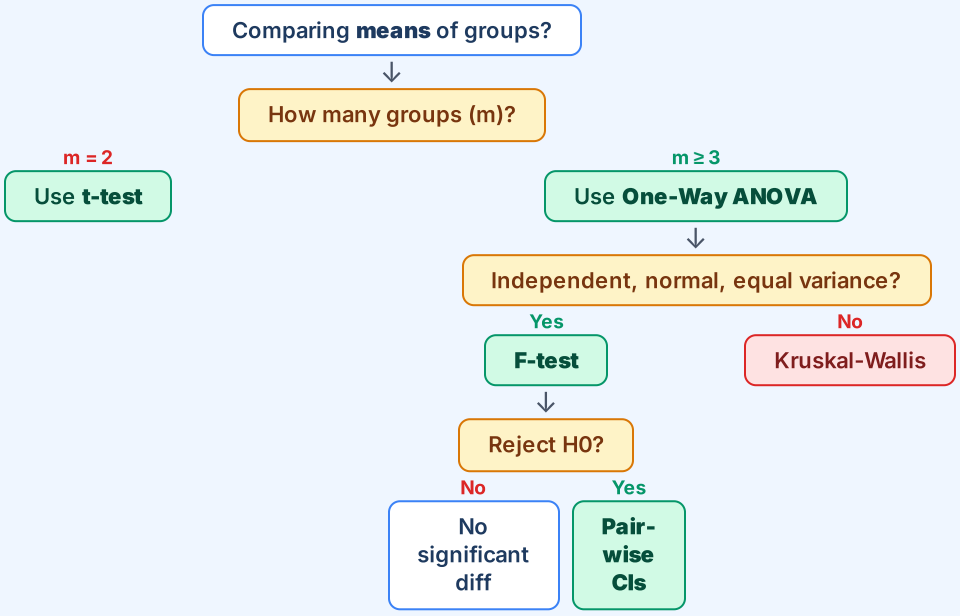
ANOVA Cheatsheet

One-Way Analysis of Variance — Formulas + When to Use

1. When to Use ANOVA — Decision Flowchart



2. Model, Hypotheses & Means

Y_{ij} = response of j-th subject in i-th group; $i = 1, \ldots, m$, $j = 1, \ldots, n$. Total: $N = mn$.

MODEL

$$Y_{ij} \sim N(\mu_i, \sigma^2) \quad \text{(independent, same variance)}$$

HYPOTHESES

$H_0 : \mu_1 = \cdots = \mu_m$ vs H_1: not all equal

MEANS

Overall: $\hat{\mu} = \bar{Y}_{\cdot\cdot} = \frac{1}{mn} \sum_i \sum_j Y_{ij}$

Group: $\hat{\mu}_i = \bar{Y}_{i\cdot} = \frac{1}{n} \sum_j Y_{ij}$

ANOVA compares **variances** to test equality of **means**. Large between-group variance relative to within-group variance implies the means differ.

3. Sums of Squares

TOTAL (SST)

$$\text{SST} = \sum_{i=1}^{m}\sum_{j=1}^{n}(Y_{ij} - \hat{\mu})^2$$

Use when: Total variation around the grand mean. Baseline quantity; equals SSB + SSW.

WITHIN-GROUP (SSW) — THE "NOISE"

$$\text{SSW} = \sum_{i=1}^{m}\sum_{j=1}^{n}(Y_{ij} - \hat{\mu}_i)^2$$

Use when: Variation within each group around its own mean. Does **not** depend on whether H_0 is true.

BETWEEN-GROUP (SSB) — THE "SIGNAL"

$$\text{SSB} = n\sum_{i=1}^{m}(\hat{\mu}_i - \hat{\mu})^2$$

Use when: Variation between group means and the grand mean. **Large** when H_0 is false.

KEY RELATIONSHIP + SHORTCUT FORMULAS

$\text{SST} = \text{SSB} + \text{SSW}$ → Use: $\text{SSW} = \text{SST} - \text{SSB}$

SSB shortcut: $\text{SSB} = n\left(\sum_i \hat{\mu}_i^2 - m\hat{\mu}^2\right)$ — use when you have group means.
SSW shortcut: $\text{SSW} = \sum_i \sum_j Y_{ij}^2 - mn\hat{\mu}^2 - \text{SSB}$ — use when you have raw data.

4. F-Test & Rejection Rule

MSW (WITHIN)

$$\text{MSW} = \frac{\text{SSW}}{mn - m}$$

Always estimates σ^2. df = $m(n-1)$.

MSB (BETWEEN)

$$\text{MSB} = \frac{\text{SSB}}{m - 1}$$

Estimates σ^2 **only under** H_0. df = $m - 1$.

Distribution theory: $\text{SSW}/\sigma^2 \sim \chi^2_{mn-m}$ (always); $\text{SSB}/\sigma^2 \sim \chi^2_{m-1}$ (under H_0 only); SSW and SSB independent.

F-STATISTIC

$$F = \frac{\text{MSB}}{\text{MSW}} = \frac{\text{SSB}/(m-1)}{\text{SSW}/(mn-m)} \sim F_{mn-m,\, m-1}$$

Use when: Testing H_0. Large F → between-group variance dominates → means differ.

REJECTION RULE (A SIGNIFICANCE LEVEL)

$$\text{Reject } H_0 \quad \text{if} \quad F > F_{mn-m,\, m-1,\, \alpha}$$

df lookup: numerator $\text{df}_1 = mn - m$ (within), denominator $\text{df}_2 = m - 1$ (between). Do not mix them up!

Intuition: Under H_0, MSB ≈ MSW so $F \approx 1$. When H_0 is false, MSB ≫ MSW so $F \gg 1$.

5. ANOVA Table Template

Fill this for every problem. F goes in the Between row only. **Check:** df and SS sums must match Total.

SOURCE	DF	SS	MS	F
Between	$m-1$	SSB	$\frac{\text{SSB}}{m-1}$	$\frac{\text{MSB}}{\text{MSW}}$
Within	$mn-m$	SSW	$\frac{\text{SSW}}{mn-m}$	—
Total	$mn-1$	SST	—	—

6. Pair-Wise Comparisons (After Rejecting H0)

Compare individual group means with confidence intervals for the difference. Use $\hat{\sigma} = \sqrt{\text{MSW}}$ and df $= mn - m$.

CI FOR MA − MB (EQUAL N)

$$(\hat{\mu}_a - \hat{\mu}_b) \pm t_{mn-m,\,\alpha/2}\sqrt{\text{MSW}}\sqrt{\frac{2}{n}}$$

CI FOR MA − MB (UNEQUAL NA, NB)

$$(\hat{\mu}_a - \hat{\mu}_b) \pm t_{mn-m,\,\alpha/2}\sqrt{\text{MSW}}\sqrt{\frac{1}{n_a}+\frac{1}{n_b}}$$

Interpreting the CI:

- CI **contains 0** → fail to reject $H_0 : \mu_a = \mu_b$ (no significant difference)
- CI **does not contain 0** → reject H_0 (significant difference exists)
- CI entirely **positive** → $\mu_a > \mu_b$; entirely **negative** → $\mu_a < \mu_b$

7. Exam Procedure — Step-by-Step

- **Step 1 — Identify:** m (groups), n (obs/group), N = mn (total)
- **Step 2 — Means:** Grand mean $\hat{\mu}$, each group mean $\hat{\mu}_i$
- **Step 3 — Sums of Squares:** SSB first (from means), then SSW = SST − SSB
- **Step 4 — ANOVA Table:** Fill in df, SS, MS, F
- **Step 5 — Decision:** Compare F to $F_{mn-m,m-1,\alpha}$
- **Step 6 — If reject:** Pair-wise CIs to find which means differ

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All Formulas at a Glance:

SSB	$n\sum_i(\hat{\mu}_i - \hat{\mu})^2$ | df $= m-1$ | MSB = SSB/$(m-1)$
SSW	$\sum_i\sum_j(Y_{ij} - \hat{\mu}_i)^2$ | df $= mn-m$ | MSW = SSW/$(mn-m)$
F	$F = \text{MSB}/\text{MSW} \sim F_{mn-m,\,m-1}$ | Reject if $F > F_{\text{crit}}$
CI	$(\hat{\mu}_a - \hat{\mu}_b) \pm t_{mn-m,\,\alpha/2}\sqrt{\text{MSW}}\sqrt{2/n}$ | df $= mn-m$